Stochastic Calculus and Option Pricing
Lecture 11 Exercises

11 Week 11 — Change of Numeraire

Focus. The exercise tests the no-arbitrage structure of currency quotes.

Exercise 11.1 (Three currency quotes must agree)**.** Let $X^{A/B}$ mean units of currency A per unit of currency B, and define $X^{B/C}$ and $X^{A/C}$ similarly.

(a) Write the spot-quote identity that rules out a triangular arbitrage. Explain the units on both sides.

(b) If the identity fails, describe the cycle of conversions that produces a positive terminal amount from one unit of currency.

(c) Explain why the arbitrage argument uses traded exchange rates directly; no choice of numeraire can repair inconsistent spot quotes.